

Exercise 2.5

Expression for the superficial degree of divergence:

in  $d$ -dimension the loop integral has  $d$  amount of momentum in the numerator

every Dirac propagator gives one  $p$  in the denominator and every photon gives  $p^2$  in the denominator so we get that the <sup>superficial</sup> Degree of divergence

is 
$$D = dL - P_e - 2P_r$$

The number of loops in a diagram is given by  $L = P_e + P_r - V + 1$

$$\begin{aligned} P_e &= \# \text{ of Dirac propagators} \\ P_r &= \# \text{ of photon propagators} \\ V &= \# \text{ of vertices} \\ N_e &= \# \text{ of external legs for Dirac particles} \\ N_r &= \# \text{ of photon external legs} \end{aligned}$$

Example:   $P_r = 2, P_e = 2, V = 4$   
 $L = 1$

  $P_r = 3, P_e = 4, V = 6, L = 3 + 4 - 6 + 1 = 2$

Every vertex has two fermion lines and a single photon line and a propagator connects to two vertices. This gives us:

$$\begin{aligned} V &= 2P_r + N_r = \frac{1}{2}(2P_e + N_e) \Rightarrow P_r = \frac{1}{2}(V - N_r) \\ &\Rightarrow P_e = (2V - N_e)\frac{1}{2} \end{aligned}$$

$$\Rightarrow L = P_e + P_r - V + 1 =$$

$$D = dP_e + dP_r - dV + d - P_e - 2P_r$$

$$= \frac{1}{2}d(2V - N_e) + \frac{1}{2}d(V - N_r) - dV + d - (2V - N_e)\frac{1}{2} - V - N_r$$

$$= V(d + \frac{1}{2}d - d - 1 - 1) - N_e(\frac{1}{2}d - \frac{1}{2}) - N_r(\frac{1}{2}d - 1) + d$$

$$D = d + V(\frac{d-4}{2}) - (\frac{d-2}{2})N_r - (\frac{d-1}{2})N_e$$

$$\Rightarrow D = d + (\frac{d-4}{2})V - (\frac{d-2}{2})N_r - (\frac{d-1}{2})N_e \quad \square$$

## Exercise 2.6

With  $d=4$  the superficial degree of divergence becomes

$$D = 4 - N_r - \frac{3}{2} N_e$$

Let's now find all 1PI-diagrams with  $D \geq 0$

- (1)  $N_r = 0$  and  $N_e = 0$  gives the first one:   $D = 4$  // Vanishes because a c.c.s. shift in vacuum energy which isn't observable
- (2)  $N_r = 1$  and  $N_e = 0$  gives   $D = 3$  // Vanishes due to symmetry
- (3)  $N_r = 0$  and  $N_e = 1$  gives   $D = \frac{5}{2}$  // This vanishes since Lorentz invariance is required
- (4)  $N_r = 2$  and  $N_e = 0$  gives   $D = 2$  // Is divergent and doesn't vanish
- (5)  $N_r = 3$  and  $N_e = 0$  gives   $D = 1$  // Vanishes due to symmetry
- (6)  $N_r = 4$  and  $N_e = 0$  gives   $D = 0$  // Ward identity cancel the divergencies
- (7)  $N_r = 0$  and  $N_e = 2$  gives   $D = 1$  // Doesn't vanish
- (8)  $N_r = 1$  and  $N_e = 2$  gives   $D = 0$  // Doesn't vanish
- (9)  $N_r = 2$  and  $N_e = 1$  gives   $D = \frac{1}{2}$  // Vanishes
- (10)  $N_r = 1$  and  $N_e = 1$  gives   $D = \frac{3}{2}$  // Vanishes

So we have 10 diagrams with non-negative superficial degree of divergence of which only 3 need to be considered for calculations for renormalization since the diagrams vanish or the divergencies get cancelled by Ward identity.

## Exercise 2.7

Compute the integral

$$B_0(m_1^2, m_2^2, p^2) = \mu^{4-d} \int \frac{d^d k}{(2\pi)^d} \frac{1}{(k^2 - m_1^2 + i\epsilon)(k-p)^2 - m_2^2 + i\epsilon}$$

In dimensional regularization

Let's use Feynman parametrization:  $\frac{1}{AB} = \int_0^1 dx \frac{1}{[xA + (1-x)B]^2}$

$$\Rightarrow B_0(m_1^2, m_2^2, p^2) \equiv B_0 = \mu^{4-d} \int \frac{d^d k}{(2\pi)^d} dx \frac{1}{[x((k-p)^2 - m_2^2 + i\epsilon) + (1-x)(k^2 - m_1^2 + i\epsilon)]^2}$$

Let's now manipulate the denominator to a desirable form:

$$\begin{aligned} D &= x[(k-p)^2 - m_2^2 + i\epsilon] + (1-x)[k^2 - m_1^2 + i\epsilon] \\ &= x(k^2 - 2k \cdot p + p^2 - m_2^2 + i\epsilon) + k^2 - m_1^2 + i\epsilon - xk^2 + m_1^2 x - xi\epsilon \\ &= x(p^2 - 2k \cdot p) + k^2 - xm_2^2 - m_1^2 + m_1^2 x + i\epsilon \\ &= k^2 - 2k \cdot (px) + p^2 x^2 - x^2 p^2 + xp^2 + x(m_1^2 - m_2^2) - m_1^2 + i\epsilon \\ &= (k - px)^2 + p^2(x - x^2) + x(m_1^2 - m_2^2) - m_1^2 + i\epsilon \\ &= l^2 - \Delta + i\epsilon \end{aligned} \quad \left\| \begin{array}{l} \text{Substitution } l = k - px \\ \Rightarrow dk \rightarrow dl \\ \Delta = -p^2(x - x^2) - x(m_1^2 - m_2^2) + m_1^2 \end{array} \right.$$

$$\Rightarrow B_0 = \mu^{4-d} \int_0^1 dx \int \frac{d^d l}{(2\pi)^d} \frac{1}{[l^2 - \Delta + i\epsilon]^2}$$

Now Let's do Wick rotation to get to the Euclidean space:

$$\begin{aligned} l^0 &\rightarrow i l_E^0 \quad \text{and} \quad l^i \rightarrow l_E^i \quad \Rightarrow \quad d^d l \rightarrow i d^d l_E \\ \text{and} \quad l^2 &\rightarrow -l_E^2 \quad \text{and} \quad i\epsilon \text{ is not needed anymore since we bypassed the poles with the rotation} \end{aligned}$$

$$\begin{aligned} \Rightarrow B_0 &= \mu^{4-d} i \int_0^1 dx \int \frac{d^d l_E}{(2\pi)^d} \frac{1}{(l_E^2 + \Delta)^2} \quad \left\| \begin{array}{l} \text{Now we change to the} \\ \text{Spherical coordinates} \quad d^d l_E = d\Omega_d dl_E l_E^{d-1} \\ \equiv d\Omega \, dl_E \, l_E^{d-1} \end{array} \right. \\ &= i \mu^{4-d} \frac{1}{(2\pi)^d} \int dx \int d\Omega \int dl_E \frac{l_E^{d-1}}{(l_E^2 + \Delta)^2} \end{aligned}$$

Let's use the results for this Euclidean integral from Peskin (A.44)

$$\Rightarrow B_0 = i \mu^{4-d} \int_0^1 dx \frac{1}{(4\pi)^{d/2}} \frac{\Gamma(2-d/2)}{\Gamma(2)} \left(\frac{1}{\Delta}\right)^{2-d/2} \quad \parallel \Gamma(2) = 1$$

$$B_0 = i \frac{\mu^{4-d}}{(4\pi)^{d/2}} \int_0^1 dx \Gamma(2-d/2) \left(\frac{1}{\Delta}\right)^{2-d/2}$$

$d = 4 - \epsilon$  in our case so we get

$$\Gamma(2 - \frac{d}{2}) = \Gamma\left(\frac{4 - 4 + \epsilon}{2}\right) = \Gamma\left(\frac{\epsilon}{2}\right), \quad (4\pi)^{d/2} = (4\pi)^2 (4\pi)^{-\epsilon/2}$$

$$\mu^{4-d} = \mu^\epsilon \quad \text{and} \quad \left(\frac{1}{\Delta}\right)^{2-d/2} = \left(\frac{1}{\Delta}\right)^{\epsilon/2}$$

$$\Rightarrow B_0 = i \left[ \frac{\mu^2}{(4\pi)} \right]^{\epsilon/2} \frac{1}{(4\pi)^2} \int_0^1 dx \Gamma\left(\frac{\epsilon}{2}\right) \left(\frac{1}{\Delta}\right)^{\epsilon/2}$$

Now we use approximations:  $\Gamma\left(\frac{\epsilon}{2}\right) = \frac{2}{\epsilon} - \gamma + \mathcal{O}(\epsilon)$

$$\left(\frac{\mu^2}{4\pi}\right)^{\epsilon/2} = \exp\left[\frac{\epsilon}{2} \ln\left(\frac{\mu^2}{4\pi}\right)\right] \approx 1 + \frac{\epsilon}{2} \ln\left[\frac{\mu^2}{4\pi}\right] + \mathcal{O}(\epsilon^2)$$

$$\Delta^{-\epsilon/2} \approx 1 - \frac{\epsilon}{2} \ln(\Delta) + \mathcal{O}(\epsilon^2)$$

$$\Rightarrow B_0 = i \frac{1}{(4\pi)^2} \int_0^1 dx \left(\frac{2}{\epsilon} - \gamma + \mathcal{O}(\epsilon)\right) \left[1 - \frac{\epsilon}{2} \ln(\Delta) + \mathcal{O}(\epsilon^2)\right] \left[1 + \frac{\epsilon}{2} \ln\left(\frac{\mu^2}{4\pi}\right) + \mathcal{O}(\epsilon^2)\right]$$

We are interested in case of  $\epsilon \rightarrow 0$  so we need not have terms of  $\mathcal{O}(\epsilon)$

$$B_0 = \frac{i}{(4\pi)^2} \int_0^1 dx \left[\frac{2}{\epsilon} - \gamma\right] \left[1 - \frac{\epsilon}{2} \ln(\Delta) + \frac{\epsilon}{2} \ln\left(\frac{\mu^2}{4\pi}\right)\right]$$

$$= \frac{i}{(4\pi)^2} \int_0^1 dx \left[\ln\left[\frac{\mu^2}{4\pi}\right] - \ln[\Delta] - \gamma + \frac{2}{\epsilon}\right] \quad \begin{array}{l} \text{divergent part} \\ \text{Let's integrate over } \frac{2}{\epsilon}, \gamma \text{ and } \ln\left(\frac{\mu^2}{4\pi}\right) \end{array}$$

$$B_0 = \frac{i}{(4\pi)^2} \left[ \frac{2}{\epsilon} - \gamma + \ln\left[\frac{\mu^2}{4\pi}\right] - \int_0^1 dx \ln(-p^2(x-x') - x(m_1^2 - m_2^2) + m_1^2) \right]$$

$$B_0 = \frac{i}{(4\pi)^2} \left[ \frac{2}{\epsilon} - \gamma + \ln\left[\frac{\mu^2}{4\pi}\right] - \int_0^1 dx \ln((1-x)m_1^2 - x(1-x)p^2 + x m_2^2) \right]$$

Exercise 2.8

$$A_0(m^2) = \int \frac{d^D k}{(2\pi)^D} \frac{1}{k^2 + m^2}$$

$$B_0(m_1^2, m_2^2, p^2) = \int \frac{d^D k}{(2\pi)^D} \frac{1}{(k^2 + m_1^2)((k-p)^2 + m_2^2)}, \quad B_\mu(m_1^2, m_2^2, p^2) = \int \frac{d^D k}{(2\pi)^D} \frac{k_\mu}{(k^2 + m_1^2)((k-p)^2 + m_2^2)}$$

$$B_{\mu\nu}(m_1^2, m_2^2, p^2) = \int \frac{d^D k}{(2\pi)^D} \frac{k_\mu k_\nu}{(k^2 + m_1^2)((k-p)^2 + m_2^2)}$$



The Lorentz structure is  $B_\mu = p_\mu B_1$  because  $B_\mu$  has only one argument with a possibility of a Lorentz structure which is  $p^2 \Rightarrow$  so  $p_\mu$  is the only Lorentz vector that can give the Lorentz structure.

And also  $B_{\mu\nu} = \underbrace{p_\mu p_\nu}_{\uparrow} B_2 + \underbrace{g_{\mu\nu}}_{\leftarrow g_{\mu\nu} \text{ is available since it is a metric}} B_3$

Here come from same argument as for  $p_\mu B_1 = B_\mu$

2) Express  $B_1$  as linear combination of  $A_0(m_1^2)$ ,  $A_0(m_2^2)$  and  $B_0$

$$B_\mu = p_\mu B_1 \Leftrightarrow \int \frac{d^D k}{(2\pi)^D} \frac{k_\mu}{(k^2 + m_1^2)((k-p)^2 + m_2^2)} = p_\mu B_1$$

Let's multiply both sides by  $p^\mu$

$$\Rightarrow p^\mu B_\mu = \int \frac{d^D k}{(2\pi)^D} \frac{p^\mu k_\mu}{(k^2 + m_1^2)((k-p)^2 + m_2^2)} = p^\mu p_\mu B_1 \quad \parallel \quad p^\mu p_\mu = p^2$$

$$p^\mu k_\mu = p \cdot k = -\frac{1}{2}(-2 p \cdot k) = -\frac{1}{2} [k^2 - k^2 - 2 p \cdot k + p^2 - p^2 + m_1^2 - m_2^2 + m_1^2 - m_1^2]$$

$$= -\frac{1}{2} [d((k-p)^2 + m_2^2) - d(p^2 + m_2^2 - m_1^2) - d(k^2 + m_1^2)]$$

$$\Rightarrow p^2 B_1 = -\frac{1}{2} \int \frac{d^D k}{(2\pi)^D} \frac{((k-p)^2 + m_2^2) - (p^2 + m_2^2 - m_1^2) - (k^2 + m_1^2)}{(k^2 + m_1^2)((k-p)^2 + m_2^2)}$$

$$p^2 B_1 = -\frac{1}{2} \left[ \underbrace{\int \frac{d^D k}{(2\pi)^D} \frac{1}{(k^2 + m_1^2)}}_{= A_0(m_1^2)} - \int \frac{d^D k}{(2\pi)^D} \frac{1}{(k-p)^2 + m_2^2} - (p^2 + m_2^2 - m_1^2) \underbrace{\int \frac{d^D k}{(2\pi)^D} \frac{1}{(k^2 + m_1^2)((k-p)^2 + m_2^2)}}_{= B_0(m_1^2, m_2^2, p^2)} \right]$$

$$B_1 = -\frac{1}{2p^2} A_0(m_1^2) + \frac{p^2 + m_2^2 - m_1^2}{2p^2} B_0 + \left[ \int \frac{d^D k}{(2\pi)^D} \frac{1}{(k-p)^2 + m_2^2} \right] \cdot \frac{1}{2p^2}$$

$$= B_0$$

On the final integral we can do the shift  $k \rightarrow k-p = \ell \Rightarrow dk = d\ell$

$$\Rightarrow \int \frac{d^D k}{(2\pi)^D} \frac{1}{(k-p)^2 + m_2^2} = \int \frac{d^D \ell}{(2\pi)^D} \frac{1}{\ell^2 + m_2^2} = A_0(m_2^2)$$

$$\Rightarrow B_1 = -\frac{1}{2p^2} A_0(m_1^2) + \frac{1}{2p^2} A_0(m_2^2) + \frac{p^2 + m_2^2 - m_1^2}{2p^2} B_0 \quad \square$$

b) Now let's find  $B_{21}$  and  $B_{22}$

$$B_{\mu\nu} = \eta_{\mu\nu} B_{21} + g_{\mu\nu} B_{22}$$

// Let's do contraction by two independent fermions

$$\begin{cases} \eta^{\mu\nu} B_{\mu\nu} = \eta^{\mu\mu} B_{21} + \eta^{\nu\nu} B_{22} \\ g^{\mu\nu} B_{\mu\nu} = \eta^{\mu\mu} B_{21} + D B_{22} \end{cases}$$

//  $g_{\mu\nu} g^{\mu\nu} = D$  where  $D$  is the dimension of spacetime

$$\Rightarrow D \eta^{\mu\nu} B_{\mu\nu} - \eta^{\mu\mu} B_{\mu\nu} g^{\mu\nu} = \eta^{\mu\mu} (D-1) B_{21} \quad \text{And}$$

$$\eta^{\mu\mu} B_{\mu\nu} g^{\mu\nu} - \eta^{\mu\nu} B_{\mu\nu} = \eta^{\mu\mu} (D-1) B_{22}$$

Let's find  $B_{21}$  first:

$$(D \eta^{\mu\nu} - \eta^{\mu\mu} g^{\mu\nu}) B_{\mu\nu} = \int \frac{d^D k}{(2\pi)^D} \frac{k_\mu k_\nu (D \eta^{\mu\nu} - \eta^{\mu\mu} g^{\mu\nu})}{(k^2 + m_1^2)((k-p)^2 + m_2^2)} = \eta^{\mu\mu} (D-1) B_{21}$$

In the loop integral the symmetry allows us to write

$$k^\mu k^\nu = \frac{1}{D} k^2 g^{\mu\nu} \quad (\text{Peskin (A.41)})$$

Using this we see that

$$(D \eta^{\mu\nu} - \eta^{\mu\mu} g^{\mu\nu}) B_{\mu\nu} = \int \frac{d^D k}{(2\pi)^D} \frac{(D k^2 g^{\mu\nu} \eta^{\mu\nu} \frac{1}{D} - \eta^{\mu\mu} k^2)}{(k^2 + m_1^2)((k-p)^2 + m_2^2)} = 0 \Rightarrow B_{21} = 0$$

So we have  $B_{21} = 0$ , Now let's compute  $B_{22}$

$$\begin{aligned} \eta^{\mu\mu} (D-1) B_{22} &= \eta^{\mu\mu} B_{\mu\nu} g^{\mu\nu} - \eta^{\mu\nu} B_{\mu\nu} = \int \frac{d^D k}{(2\pi)^D} \frac{k_\mu (p^2 g^{\mu\nu} - \eta^{\mu\nu} p^2)}{(k^2 + m_1^2)((k-p)^2 + m_2^2)} \\ &= \int \frac{d^D k}{(2\pi)^D} \frac{p^2 (1 - \frac{1}{D}) k^2}{(k^2 + m_1^2)((k-p)^2 + m_2^2)} = p^2 (1 - \frac{1}{D}) \int \frac{d^D k}{(2\pi)^D} \frac{(k^2 + m_1^2) - m_1^2}{(k^2 + m_1^2)((k-p)^2 + m_2^2)} \end{aligned}$$

$$\eta^{\mu\mu} (D-1) B_{22} = p^2 \frac{1}{D} (D-1) [A_0(m_1^2) - m_1^2 B_0]$$

$$// A_0(m_1^2) = \int \frac{d^D k}{(2\pi)^D} \frac{1}{(k-p)^2 + m_2^2}$$

since we can do a shift  $k-p=l$  and  $dl=dk$

$$\Rightarrow B_{22} = \frac{1}{D} A_0(m_1^2) - \frac{m_1^2}{D} B_0 \quad \square$$

And  $B_{21} = 0$